Aramco summer course application tracker

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Introduction

Project Idea:

Our idea is that we are creating very basic databases system that will help:

1. Efficient Management:
Simplify the process of managing student applications for Aramco summer courses.

2. Centralized Storage:
Store all application details in one digital database for easy access and organization.

3. Improved Communication:
Facilitate communication between administrators and applicants through automated notifications.

4. Streamlined Process:
Enhance efficiency and accuracy by automating tasks and reducing manual paperwork.

Problem description:

The problem that faced Aramco company was the need for an efficient and organized system to manage their summer program. The existing manual processes and scattered information made it difficult to track and monitor the program's progress, including participant details. Therefore, Aramco decided to create a simple database system to streamline their summer program management and enhance overall efficiency.

Project Goals:

1. Efficient Management:
Simplify the process of managing student applications for Aramco summer courses.

2. Centralized Storage:
Store all application details in one digital database for easy access and organization.

3. Improved Communication:
Facilitate communication between administrators and applicants through automated notifications.

4. Streamlined Process:
Enhance efficiency and accuracy by automating tasks and reducing manual paperwork.

Phase 1

List of Entities:

Entities	Attributes
Student	student_id, name, email, phone, national_id, high_school_name, current_grade, current_gpa, place_of_birth, date_of_birth, gender
Program	program_id, name, description
Admin	admin_id, name, email, phone

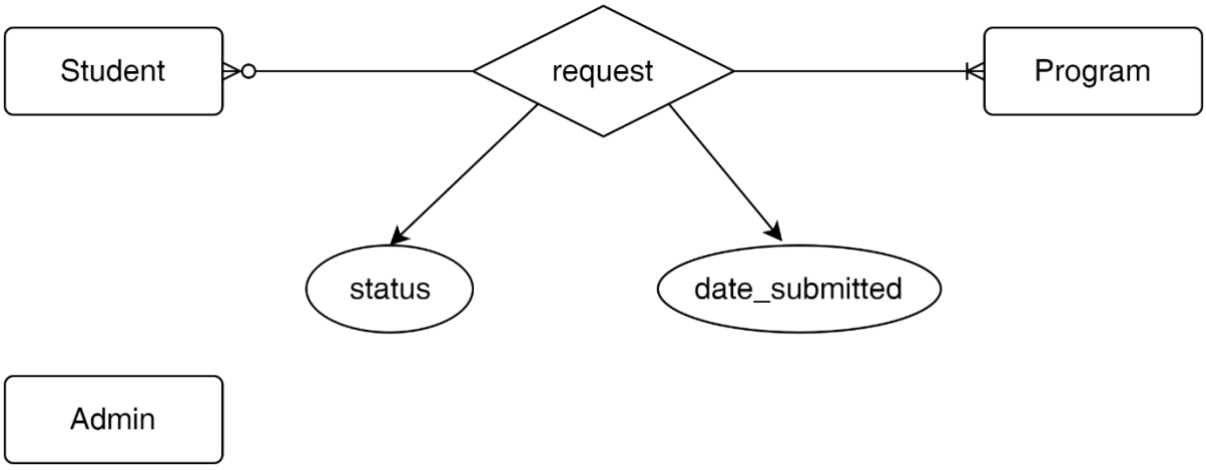
Relationships:

<u>Student, Program:</u>

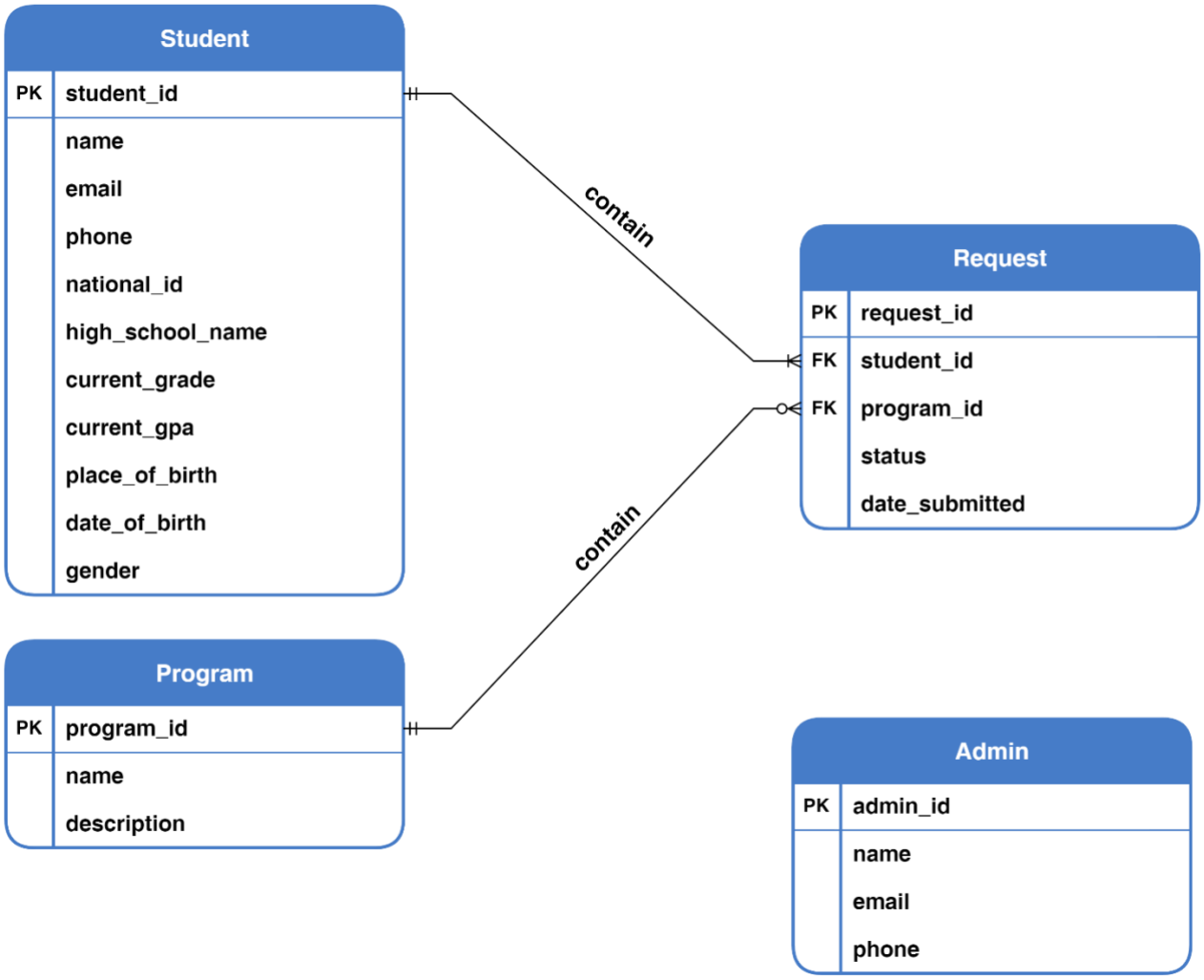
A student can submit a request for one or more programs, and the program can also be requested by several students.

(Many-to-Many Relationship)

Conceptual ER:



ER Diagram (ERD):



Phase 2

Relational Schema:

Student

student_id	name	email	phone	national_id	high_school_name	current_grade	
							current_gpa	place_of_birth	date_of_birth	gender

Request

FK FK

request_id	student_id	program_id	status	date_submitted

Program

program_id	name	description

Admin

admin_id	name	email	phone

Functional dependencies (FD):

student_id → name, email, phone, national_id, high_school_name, current_grade, current_gpa, place_of_birth, date_of_birth, gender.

program_id → name, description.

request_id → student_id, program_id, status, date_submitted.

admin_id → name, email, phone.

Normalization:

1NF (First Normal Form): A table is in 1NF if and only if there are no repeating groups or multivalued attributes.

1- There are no repeating groups.
2- No composite attributes.

The schema is already in 1NF.

2NF (Second Normal Form): A table is in 2NF if and only if it is in 1NF and every non-key attribute is fully functional and dependent on the primary key.

3- All tables are in 1NF.
4- There is no partial functional dependency.

The schema is already in 2NF.

3NF (Third Normal Form): A table is in 3NF if and only if it is in 2NF and there are no transitive dependencies between attributes.

5- All tables are in 2NF.
6- There is no transitive functional dependency.

The schema is already in 3NF.

Phase 3

Table Name	Attributes	Data Type	Constraints
Student	1- student_id 2- name 3- email 4- phone 5- national_id 6- high_school_name 7- current_grade 8- current_gpa 9- place_of_birth 10- date_of_birth 11- gender	1- Integer 2- Varchar (255) 3- Varchar (255) 4- Varchar (15) 5- Varchar (255) 6- Varchar (255) 7- Varchar (100) 8- Decimal (4, 2) 9- Varchar (255) 10- Date 11- Varchar (6)	PRIMARY KEY CONSTRAINT gender_check CHECK (gender IN ('Male', 'Female'))
Program	1- program_id 2- name 3- description	1- Integer 2- Varchar (255) 3- Varchar (1000)	PRIMARY KEY
Request	1- request_id 2- student_id 3- program_id 4- status 5- date_submitted	1- Integer 2- Integer 3- Integer 4- Varchar (255) 5- Date	PRIMARY KEY FOREIGN KEY FOREIGN KEY
Admin	1- admin_id 2- name 3- email 4- phone	1- Integer 2- Varchar (255) 3- Varchar (255) 4- Varchar (15)	PRIMARY KEY

Tables:

Student Table:

```sql
CREATE TABLE Student (
    student_id INT PRIMARY KEY,
    name VARCHAR(255),
    email VARCHAR(255),
    phone VARCHAR(15),
    national_id VARCHAR(255),
    high_school_name VARCHAR(255),
    current_grade VARCHAR(100),
    current_gpa DECIMAL(4, 2),
    place_of_birth VARCHAR(255),
    date_of_birth DATE,
    gender VARCHAR(6),
    CONSTRAINT gender_check CHECK (gender IN ('Male', 'Female'))
);
```

```
Table created.
```

Insert data into Student Table:

```sql
INSERT INTO Student (student_id, name, email, phone, national_id, high_school_name, current_grade, current_gpa, place_of_birth, date_of_birth, gender)
VALUES (1, 'Ahmed Ali', 'ahmed@example.com', '0501234567', '1234567890', 'King Fahd Secondary School', 'Grade 10', 7.5, 'Riyadh', TO_DATE('2005-05-15', 'YYYY-MM-DD'), 'Male');

INSERT INTO Student (student_id, name, email, phone, national_id, high_school_name, current_grade, current_gpa, place_of_birth, date_of_birth, gender)
VALUES (2, 'Fatima Mohammed', 'fatima@example.com', '0549876543', '0987654321', 'King Abdulaziz Secondary School', 'Grade 11', 8.5, 'Jeddah', TO_DATE('2006-07-20', 'YYYY-MM-DD'), 'Female');

INSERT INTO Student (student_id, name, email, phone, national_id, high_school_name, current_grade, current_gpa, place_of_birth, date_of_birth, gender)
VALUES (3, 'Ali Ahmed', 'ali@example.com', '0555555555', '11111111', 'King Khalid Secondary School', 'Grade 12', 7.0, 'Dammam', TO_DATE('2007-10-10', 'YYYY-MM-DD'), 'Male');

INSERT INTO Student (student_id, name, email, phone, national_id, high_school_name, current_grade, current_gpa, place_of_birth, date_of_birth, gender)
VALUES (4, 'Sara Abdullah', 'sara@example.com', '0577777777', '22222222', 'King Fahd Secondary School', 'Grade 10', 9.5, 'Medina', TO_DATE('2005-01-05', 'YYYY-MM-DD'), 'Female');

INSERT INTO Student (student_id, name, email, phone, national_id, high_school_name, current_grade, current_gpa, place_of_birth, date_of_birth, gender)
VALUES (5, 'Omar Khaled', 'omar@example.com', '0599999999', '33333333', 'King Abdulaziz Secondary School', 'Grade 11', 8.0, 'Taif', TO_DATE('2006-03-25', 'YYYY-MM-DD'), 'Male');

INSERT INTO Student (student_id, name, email, phone, national_id, high_school_name, current_grade, current_gpa, place_of_birth, date_of_birth, gender)
VALUES (6, 'Layla Hassan', 'layla@example.com', '0506666666', '44444444', 'King Khalid Secondary School', 'Grade 12', 7.5, 'Abha', TO_DATE('2007-09-08', 'YYYY-MM-DD'), 'Female');

INSERT INTO Student (student_id, name, email, phone, national_id, high_school_name, current_grade, current_gpa, place_of_birth, date_of_birth, gender)
VALUES (7, 'Khalid Saleh', 'khalid@example.com', '0508888888', '55555555', 'King Fahd Secondary School', 'Grade 10', 8.5, 'Qassim', TO_DATE('2005-11-30', 'YYYY-MM-DD'), 'Male');
```

```sql
SELECT * FROM Student;
```

STUDENT_ID	NAME	EMAIL	PHONE	NATIONAL_ID	HIGH_SCHOOL_NAME	CURRENT_GRADE	CURRENT_GPA	PLACE_OF_BIRTH	DATE_OF_BIRTH	GENDER
1	Ahmed Ali	ahmed@example.com	0501234567	1234567890	King Fahd Secondary School	Grade 10	7.5	Riyadh	15-MAY-05	Male
2	Fatima Mohammed	fatima@example.com	0549876543	0987654321	King Abdulaziz Secondary School	Grade 11	8.5	Jeddah	20-JUL-06	Female
3	Ali Ahmed	ali@example.com	0555555555	11111111	King Khalid Secondary School	Grade 12	7	Dammam	10-OCT-07	Male
4	Sara Abdullah	sara@example.com	0577777777	22222222	King Fahd Secondary School	Grade 10	9.5	Medina	05-JAN-05	Female
5	Omar Khaled	omar@example.com	0599999999	33333333	King Abdulaziz Secondary School	Grade 11	8	Taif	25-MAR-06	Male
6	Layla Hassan	layla@example.com	0506666666	44444444	King Khalid Secondary School	Grade 12	7.5	Abha	08-SEP-07	Female
7	Khalid Saleh	khalid@example.com	0508888888	55555555	King Fahd Secondary School	Grade 10	8.5	Qassim	30-NOV-05	Male

Program Table:

```sql
CREATE TABLE Program (
    program_id INT PRIMARY KEY,
    name VARCHAR(255),
    description VARCHAR(1000)
);
```

Table created.

Insert data into Program Table:

```sql
INSERT INTO Program (program_id, name, description) VALUES (1, 'Summer Training Program',
    'The Summer Training Program is held during the summer break and targets high school students.
    It offers participants subjects in English to support their academic development, along with life skills and personal development with
    a focus on safety principles.');

INSERT INTO Program (program_id, name, description) VALUES (2, 'Ambition Program',
    'The Ambition Program targets outstanding high school students from various cities in the kingdom, aiming to develop students skills
    and enhance their ambitions.');

INSERT INTO Program (program_id, name, description) VALUES (3, 'Progression Program for Secondary School',
    'This program assists participants in acquiring the necessary skills for operational, administrative, clerical, and maintenance jobs.');

INSERT INTO Program (program_id, name, description) VALUES (4, 'University Follow-up Program',
    'Scholarship program for Saudi students.');

INSERT INTO Program (program_id, name, description) VALUES (5, 'University Scholarship Program for High School Graduates',
    'A carefully selected scholarship program for Saudi high school graduates of both genders.');
```

```sql
SELECT * FROM Program;
```

PROGRAM_ID	NAME	DESCRIPTION
1	Summer Training Program	The Summer Training Program is held during the summer break and targets high school students. It offers participants subjects in English to support their academic development, along with life skills and personal development with a focus on safety principles.
2	Ambition Program	The Ambition Program targets outstanding high school students from various cities in the kingdom, aiming to develop students skills and enhance their ambitions.
3	Progression Program for Secondary School	This program assists participants in acquiring the necessary skills for operational, administrative, clerical, and maintenance jobs.
4	University Follow-up Program	Scholarship program for Saudi students.
5	University Scholarship Program for High School Graduates	A carefully selected scholarship program for Saudi high school graduates of both genders.

Request Table:

```sql
CREATE TABLE Request (
    request_id INT PRIMARY KEY,
    student_id INT,
    program_id INT,
    status VARCHAR(50),
    date_submitted DATE,
    FOREIGN KEY (student_id) REFERENCES Student(student_id),
    FOREIGN KEY (program_id) REFERENCES Program(program_id)
);
```

Table created.

Insert data into Program Table:

```sql
INSERT INTO Request (request_id, student_id, program_id, status, date_submitted)
    VALUES (1, 1, 1, 'Pending', TO_DATE('2024-04-14', 'YYYY-MM-DD'));

INSERT INTO Request VALUES (2, 2, 1, 'Pending', TO_DATE('2024-04-14', 'YYYY-MM-DD'));

INSERT INTO Request VALUES (3, 3, 2, 'Pending', TO_DATE('2024-04-13', 'YYYY-MM-DD'));

INSERT INTO Request VALUES (4, 4, 2, 'Pending', TO_DATE('2024-04-13', 'YYYY-MM-DD'));

INSERT INTO Request VALUES (5, 5, 1, 'Pending', TO_DATE('2024-04-12', 'YYYY-MM-DD'));

INSERT INTO Request VALUES (6, 6, 1, 'Pending', TO_DATE('2024-04-12', 'YYYY-MM-DD'));

INSERT INTO Request VALUES (7, 7, 1, 'Pending', TO_DATE('2024-04-12', 'YYYY-MM-DD'));
```

```sql
SELECT * FROM Request;
```

REQUEST_ID	STUDENT_ID	PROGRAM_ID	STATUS	DATE_SUBMITTED
1	1	1	Pending	14-APR-24
2	2	1	Pending	14-APR-24
3	3	2	Pending	13-APR-24
4	4	2	Pending	13-APR-24
5	5	1	Pending	12-APR-24
6	6	1	Pending	12-APR-24
7	7	1	Pending	12-APR-24

Admin Table:

```sql
CREATE TABLE Admin (
    admin_id INT PRIMARY KEY,
    name VARCHAR(255),
    email VARCHAR(255),
    phone VARCHAR(15)
);
```

Table created.

Insert data into Admin Table:

```sql
INSERT INTO Admin (admin_id, name, email, phone)
    VALUES (1, 'Ahmed Ali', 'ahmed@example.com', '0501122334');

INSERT INTO Admin VALUES (2, 'Sara Jamal', 'sarah@example.com', '0504433221');

INSERT INTO Admin VALUES (3, 'Mohammed Hassan', 'mohammed@example.com', '0507788990');

INSERT INTO Admin VALUES (4, 'Fatima Ahmed', 'fatima@example.com', '0509988776');

INSERT INTO Admin VALUES (5, 'Ali Khaled', 'ali@example.com', '0501122334');

INSERT INTO Admin VALUES (6, 'Nour Abdullah', 'nour@example.com', '0502233445');
```

```sql
SELECT * FROM Admin;
```

ADMIN_ID	NAME	EMAIL	PHONE
1	Ahmed Ali	ahmed@example.com	0501122334
2	Sara Jamal	sarah@example.com	0504433221
3	Mohammed Hassan	mohammed@example.com	0507788990
4	Fatima Ahmed	fatima@example.com	0509988776
5	Ali Khaled	ali@example.com	0501122334
6	Nour Abdullah	nour@example.com	0502233445

Design Queries:

1- Data Retrieval with SELECT, JOIN, ORDER BY and WHERE:

Retrieve students who have applied to join the summer program and sort them by submitted_date field

```sql
SELECT Student.student_id, Student.name, Request.status, Request.date_submitted FROM Request
    JOIN Student ON Student.student_id = Request.student_id
    WHERE Request.program_id = 1
    ORDER BY Request.date_submitted DESC;
```

STUDENT_ID	NAME	STATUS	DATE_SUBMITTED
2	Fatima Mohammed	Pending	14-APR-24
1	Ahmed Ali	Pending	14-APR-24
6	Layla Hassan	Pending	12-APR-24
7	Khalid Saleh	Pending	12-APR-24
5	Omar Khaled	Pending	12-APR-24

2- Data Retrieval with SELECT, JOIN, GROUP BY and AGGREGATE FUNCTIONS:

Retrieve the number of students applying for programs.

```sql
SELECT Program.program_id, Program.name AS program_name, COUNT(Request.program_id) AS requests_count
FROM Program
LEFT JOIN Request ON Program.program_id = Request .program_id
GROUP BY Program.program_id, Program. name;
```

PROGRAM_ID	PROGRAM_NAME	REQUESTS_COUNT
1	Summer Training Program	5
2	Ambition Program	2
4	University Follow-up Program	0
5	University Scholarship Program for High School Graduates	0
3	Progression Program for Secondary School	0

3- Data Retrieval with SELECT, SUBQUERY and AGGREGATE FUNCTIONS:

Know and retrieve program data most requested by students.

```sql
SELECT * FROM Program WHERE program_id =
    (SELECT program_id FROM (
    SELECT program_id, COUNT(*) FROM Request GROUP BY program_id ORDER BY COUNT(*) DESC
    ) WHERE ROWNUM = 1
    );
```

PROGRAM_ID	NAME	DESCRIPTION
1	Summer Training Program	The Summer Training Program is held during the summer break and targets high school students. It offers participants subjects in English to support their academic development, along with life skills and personal development with a focus on safety principles.

4- Data Retrieval with SELECT, SUBQUERY and AGGREGATE FUNCTIONS:

Retrieving students whose average is greater than or equal to 8 and who are in the second year of secondary school.

```sql
SELECT student_id, name, email, phone, current_grade, current_gpa, gender FROM Student
    WHERE current_grade = 'Grade 11' AND current_gpa >= 8;
```

STUDENT_ID	NAME	EMAIL	PHONE	CURRENT_GRADE	CURRENT_GPA	GENDER
2	Fatima Mohammed	fatima@example.com	0549876543	Grade 11	8.5	Female
5	Omar Khaled	omar@example.com	0599999999	Grade 11	8	Male

Stored Procedures:

1- Stored Procedure using a cursor to retrieve the students of the program:

```
CREATE PROCEDURE GetApplicantsForProgram(
    p_program_id INT
)
AS
    CURSOR applicants_cur IS
        SELECT Student.student_id, Student.name, Student.email, Student.current_gpa
        FROM Student, Request
        WHERE Student.student_id = Request.student_id
        AND Request.program_id = p_program_id;
BEGIN
    FOR s_record IN applicants_cur
    LOOP
        DBMS_OUTPUT.PUT_LINE('-- Student id: ' || s_record.student_id ||
            '   Student name: ' || s_record.name || '    Student email: ' ||
            s_record.email || '     Student current gpa: ' || s_record.current_gpa);
    END LOOP;
END;
```

```
Procedure created.
```

Execute:

```
EXECUTE GetApplicantsForProgram(2);
```

```
Statement processed.
-- Student id: 3        Student name: Ali Ahmed  Student email: ali@example.com         Student current gpa: 7
-- Student id: 4        Student name: Sara Abdullah      Student email: sara@example.com                Student current gpa: 9.5
```

2- Update query based stored procedure to update program description:

```
CREATE PROCEDURE UpdateProgramDescription(
    p_program_id INT,
    new_description VARCHAR
)
AS
BEGIN
    UPDATE Program
    SET description = new_description
    WHERE program_id = p_program_id;

    COMMIT;
    DBMS_OUTPUT.PUT_LINE('Program description updated successfully.');
END;
```

```
Procedure created.
```

Execute:

```
EXECUTE UpdateProgramDescription(4, 'Scholarship Program for Saudi Male and Female Students')
```

```
Statement processed.
Program description updated successfully.
```

```
SELECT * FROM Program;
```

PROGRAM_ID	NAME	DESCRIPTION
1	Summer Training Program	The Summer Training Program is held during the summer break and targets high school students. It offers participants subjects in English to support their academic development, along with life skills and personal development with a focus on safety principles.
2	Ambition Program	The Ambition Program targets outstanding high school students from various cities in the kingdom, aiming to develop students skills and enhance their ambitions.
3	Progression Program for Secondary School	This program assists participants in acquiring the necessary skills for operational, administrative, clerical, and maintenance jobs.
4	University Follow-up Program	Scholarship Program for Saudi Male and Female Students
5	University Scholarship Program for High School Graduates	A carefully selected scholarship program for Saudi high school graduates of both genders.